

Day 12 Task Report: Webhooks and Voice Agent Integration

November 13, 2025

1 Understanding Webhooks

Webhooks are a fundamental concept for connecting n8n to external services. They are often confused with API requests, but they serve the opposite function.

- **API Request vs. Webhook:** In a typical API request, n8n *sends* a request to an external service (e.g., "Get my data from Google Sheets"). With a webhook, n8n *is the server*. It provides a unique URL and *waits* for an external service to send data *to it*.
- **The n8n Webhook Node:** This node is a **trigger**. When you add it to a workflow, it generates a unique URL. Your goal is to provide this URL to an external service (like 11Labs, Stripe, or a custom form) and configure that service to send data to this address.
- **Method Alignment:** The HTTP method must match. If the external service is sending data (which it almost always is), it will use a POST request. You must set the "HTTP Method" on your n8n Webhook node to POST to match it.

1.1 Critical Note: Test vs. Production URLs

This is the most common point of failure when using webhooks.

- **Test URL:** When you are building your workflow, you must click "**Listen for test event**". This activates the *Test URL*. You use this URL in your external service to send test data, allowing you to build the rest of your workflow.
- **Production URL:** Once you are finished building, you **must activate the workflow** (toggle it "Active"). When active, the workflow no longer listens on the Test URL. You must go back to the Webhook node, copy the *Production URL*, and paste this new URL into your external service's configuration. If you fail to update the URL, your live workflow will never receive any data.

1.2 Responding to Webhooks

Webhooks are not just for receiving data; they can also send a response back.

1. In the Webhook node, change the "Response" setting from "Respond Immediately" to "**Using 'Respond to Webhook' Node**".
2. This allows you to receive data, perform complex actions (like running an AI agent, querying a database, etc.), and then, at the end of your workflow, add a "**Respond to Webhook**" node.
3. The data you put into this "Respond" node (e.g., a confirmation message) is sent back to the original service that made the request.

2 Application: The Voice Email Agent (11Labs)

This workflow is a perfect example of using webhooks for both receiving data and sending a response.

2.1 General Architecture

The flow of conversation and data is a full circle:

1. **User → 11Labs:** The user speaks to the 11Labs voice agent (named "Archer").
2. **11Labs → n8n (Webhook):** 11Labs's internal LLM extracts key information from the user's speech (e.g., the recipient's name and the email content). It then sends this data as JSON to the n8n Webhook URL.
3. **n8n (Workflow):** The n8n workflow receives the JSON. An AI Agent node takes this data, uses its tools (Google Sheets to find the email address, Gmail to send the email), and performs the action.
4. **n8n → 11Labs (Webhook Response):** The AI Agent generates a confirmation message (e.g., "Email sent to Phil"). This message is passed to the "Respond to Webhook" node, which sends it back to 11Labs.
5. **11Labs → User:** 11Labs receives the confirmation message from n8n and speaks it back to the user in Archer's voice.

2.2 Critical Configuration: 11Labs Tool Setup

Inside the 11Labs agent setup, you must create a "Tool" (which you can name "n8n"):

- **URL:** This is where you paste your n8n Webhook URL (Test or Production).
- **Method:** Set to POST.
- **Body Parameters:** This is the most important part. You must define the JSON schema that 11Labs will send to n8n. In this example, two parameters (keys) were created:
 - **to:** A string for "who the email is going to."
 - **email_content:** A string for "what the email is about."
- **11Labs System Prompt:** The system prompt for the 11Labs agent must then be instructed to use this "n8n tool" and to fill in these specific parameters (e.g., "When the user wants to send an email, use the n8n tool and provide the 'to' and 'email_content' parameters.").

2.3 Critical Configuration: n8n Workflow

- **Webhook Trigger:** Receives the POST request from 11Labs. The data (e.g., {"to": "Phil", "email_content": "..."}) is available in `{{ $json.body }}`.
- **AI Agent:** The "User Message" for the agent is mapped to the incoming JSON from the webhook (e.g., "The email is for: `{{ $json.body.to }}`...").
- **Respond to Webhook Node:** This node takes the final output from the AI Agent (e.g., `{{ $json.output }}`) and sends it back to 11Labs, completing the loop.